

Goals: how to Eulerize a graph; why you would Eulerize a graph; how to efficiently Eulerize a graph.

1. Given a graph, when can you find:

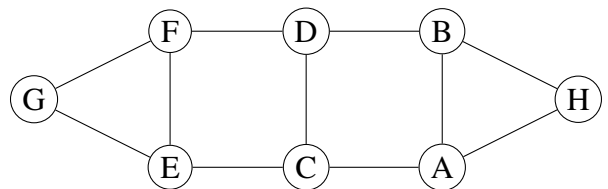
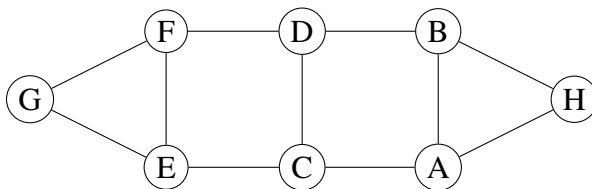
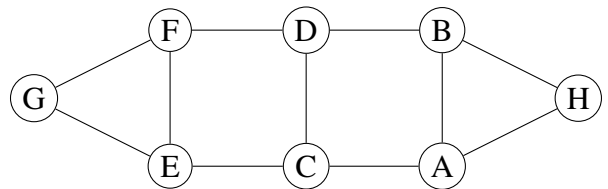
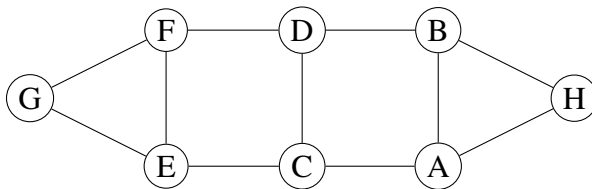
(a) An Euler circuit?

(b) An Euler path?

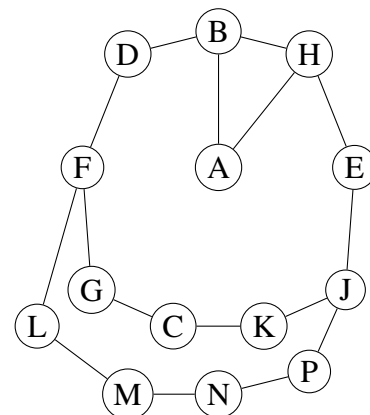
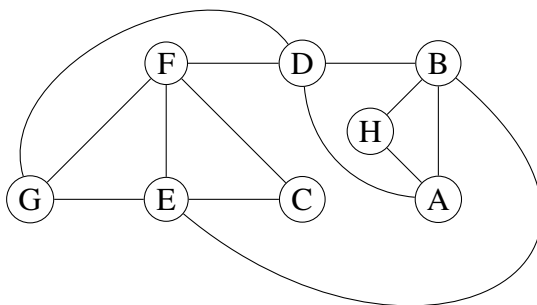
(c) Neither?

2. Consider the following graph. Why doesn't it have an Euler circuit?

Duplicate some of the edges so that every vertex is even degree. Using your additional edges, find an Euler circuit. What is the **smallest** number of edges you need to duplicate?



3. Some other examples



4. **Definition:** To **eulerize** a graph G means

5. **Definition:** An **optimal eulerization** means

6. Under what conditions do you think it is *easy* to obtain an optimal eulerization?